

OVERVIEW

As a Solutions Architect at NVIDIA (since February 2026), I work with Japanese enterprise customers on large language model training and inference, with a focus on the NeMo framework.

Previously, I served as the tech lead for LLM applications at Preferred Networks, Inc., delivering solutions for enterprise clients in sectors such as finance, healthcare, and government. My work involved customizing Large Language Models (LLMs) through techniques like continued pre-training, instruction tuning, and Direct Preference Optimization (DPO) to meet specific domain and compliance requirements.

I led a research project to build a Japanese medical domain-specialized LLM, resulting in **Llama3-Preferred-MedSwallow-70B** and **Preferred-MedLLM-Qwen-72B (paper)**, the first open LLMs that surpassed GPT-4 and GPT-4o, respectively, in the Japanese Medical Licensing Exam (JMLE). From 2021 to 2024, I led a project developing a deep learning-based solution to enhance endometriosis diagnosis from MRIs, in collaboration with a leading pharmaceutical company.

I have experience formulating problems through client consultations and providing deep learning/machine learning-based solutions. Additionally, I have mentored four research interns and two part-time engineers.

During my PhD, I developed machine learning methods to analyze gene expression/mutation data to tackle the problem of antibiotic resistance in bacteria.

EXPERIENCE

- **Solutions Architect, NVIDIA** Tokyo, Japan
Supporting Japanese enterprise customers on LLM training and inference (NeMo). Feb. 2026 - present
- **Tech Lead, Preferred Networks, Inc.** Tokyo, Japan
Tech Lead for LLM based applications for enterprise clients. Mar. 2025 - Jan. 2026
 - Led the development and customization of domain-specialized Large Language Models based on the company's flagship models, including **PLaMo**, for multiple clients.
 - * Led a project for a major Japanese securities firm to align a flagship LLM with internal compliance regulations using Supervised Fine-Tuning (SFT).
 - * Led the development of a domain-specialized LLM for a government agency through continued pre-training and SFT on a specialized domain.
 - * Developed an LLM-powered system to streamline bioinformatics analysis for a pharmaceutical company.
- **Software Engineer, Preferred Networks, Inc.** Tokyo, Japan
Developing machine learning based solutions with clients in the healthcare domain. Apr. 2021 - Mar. 2025
 - Led the development of Large Language Models (LLMs) for medical domains (Project Lead, Apr. 2024 – Mar. 2025).
 - * Led and conducted main experiments for building **Preferred-MedLLM-Qwen-72B (blog)** and **Llama3-Preferred-MedSwallow-70B (blog)**, the first open LLMs that exceeded GPT-4o and GPT-4, respectively, in the Japanese Medical Licensing Exam. Related Paper: [arXiv:2504.18080](https://arxiv.org/abs/2504.18080).
 - * Led a comparative study of RAG and continued pretraining for medical domains: [Blog link](#)
 - Developed a deep learning based solution for medical image analysis in collaboration with a pharmaceutical company, (project lead, Apr. 2021 – Sep. 2024). Related paper: [JSAI2025](#).
 - Main mentor for four interns and three part-time engineers on R&D tasks.
Related research papers/blogs: [arXiv:2506.14919](https://arxiv.org/abs/2506.14919), [arXiv:2504.18080](https://arxiv.org/abs/2504.18080), [Blog #1 \(2023\)](#), [Blog #2 \(2022\)](#).

SELECTED TALKS

1. **Junichiro Iwasawa.** “Finetuning Foundation Models for the Development and Application of Healthcare-Specific Models”. Invited talk at “Symposium: A New Era of Healthcare Opened by AI and Data Science”. Institute of Science Tokyo. March 25, 2025.
2. **Junichiro Iwasawa.** “Finetuning or RAG? Towards Building a Medical Domain-Specialized LLM”. Invited talk at “55th meeting of IBISML”. Hokkaido University. December 20, 2024.

SELECTED PUBLICATIONS

Google Scholar Profile

1. Xinkai Zhao, Yuta Tokuoka, **Junichiro Iwasawa**, Keita Oda. “Frequency-Calibrated Membership Inference Attacks on Medical Image Diffusion Models”, in *MICCAI 2025 Workshop on Deep Generative Models* (Best Paper Award), arXiv:2506.14919.
2. Wataru Kawakami, Keita Suzuki, **Junichiro Iwasawa**. “Stabilizing Reasoning in Medical LLMs with Continued Pretraining and Reasoning Preference Optimization”, Accepted to *NeurIPS 2025 Workshop on GenAI for Health*, arXiv:2504.18080.
3. **Junichiro Iwasawa**, Tomoya Maeda, Atsushi Shibai, Hazuki Kotani, Masako Kawada, Chikara Furusawa. “Analysis of the evolution of resistance to multiple antibiotics enables prediction of the *Escherichia coli* phenotype-based fitness landscape”, *PLoS Biology* **20**, 12 e3001920 (2022). [Press release](#).
4. Tomoya Maeda*, **Junichiro Iwasawa***, Hazuki Kotani, Natsue Sakata, Masako Kawada, Takaaki Horinouchi, Aki Sakai, Kumi Tanabe, and Chikara Furusawa (*co-first authors), “High-throughput laboratory evolution reveals evolutionary constraints in *Escherichia coli*”, *Nature Communications* **11**, 5970 (2020). [Press Release](#) & Recommended in [Faculty Opinions](#) as an exceptional paper.
5. **Junichiro Iwasawa**, Yuichiro Hirano, Yohei Sugawara, “Label-Efficient Segmentation using Semi-Supervised Multi-Task Learning”, In Brain Lesion (BrainLes) Workshop at *Medical Image Computing & Computer Assisted Intervention (MICCAI)* 2020. arXiv:2009.11160.

EDUCATION

- **Dept. of Physics, The University of Tokyo** Tokyo, Japan
Ph.D. in Physics Apr. 2018 - Mar. 2021
 - **Research field:** Biophysics, Machine Learning w/ biological data, Drug resistance of bacteria
- **Dept. of Physics, The University of Tokyo** Tokyo, Japan
M.S. in Physics Apr. 2016 - Mar. 2018
 - **Research field:** Non-equilibrium statistical physics, Soft active matter physics
- **Research Intern, Princeton University** NJ, USA
Investigation of 1D core-collapse supernovae models, advised by Prof. Adam Burrows Sep. 2016 - Nov. 2016
- **The University of Tokyo** Tokyo, Japan
B.S. in Physics Apr. 2012 - Mar. 2016

AWARDS AND GRANTS

- **Dissertation Award, Graduate School of Science, The University of Tokyo** Dissertation: “Deciphering evolutionary constraints through laboratory evolution and machine learning”, Mar. 2021
- **Research Fellowship for Young Scientists (DC1), Japan Society for the Promotion of Science:** Research of “Spontaneous Patterns and Order in Active Matter”, ¥2,800,000, Apr. 2018 - Mar. 2021

TECHNICAL SKILLS

- Expert: Python (PyTorch, Transformers, TRL, NumPy, Pandas, scikit-learn, etc.)
- Proficient: Kubernetes, Docker, Argo Workflows, git, LaTeX
- Familiar C++, R, FORTRAN

LINKS

- Twitter: @jiwasawa, GitHub: jiwasawa, LinkedIn: junichiro-iwasawa-875b37130, ORCID: 0000-0002-2560-5650
- Blog: <https://jiwasawa.github.io/blog-jp/> (Jp) / <https://jiwasawa.github.io/blog/> (En)